



Green AI – Eco-design of Artificial Intelligence

Balancing innovation, performance, and environmental responsibility in the age of artificial intelligence.

Why Green AI?

The Scale Challenge

Modern AI relies on **LLM – Large Language Models** (models with billions of parameters) using **GPU/TPU** (processors optimised for parallel computing / AI), resulting in massive energy consumption.

Current Impact

In 2024, AI consumes electricity equivalent to an industrialised country. Without optimisation, this will multiply by 10 by 2030.





The Innovation Paradox

Benefits

- Rapid medical diagnostics
- Process automation
- Resource optimisation
- Enhanced decision-making capabilities

Hidden Costs

- Energy-intensive data centres
- Cooling systems (**PUE**)
- Specialised hardware requirements
- Emissions from **training** and **inference** (model usage)

Key Terms in AI Energy Consumption

Parameters

Values learnt by the model during training, determining its capabilities and complexity.

Inference

Generation of responses when the model is actively used to process queries.

Cluster

Set of servers operating in parallel to handle computational workloads.

Scaling Laws

Performance increases with model size, but energy consumption rises proportionally.

Algorithmic Efficiency

Optimising AI models to reduce computational requirements whilst maintaining performance.

01

Pruning (Trimming)

Remove unnecessary parameters that contribute minimally to model performance, streamlining the architecture.

02

Quantization (Quantification)

Convert FP32 → INT8, reducing computational load and energy consumption through lower precision.

03

Distillation

Train smaller models to imitate larger ones, achieving comparable results with significantly less energy.

50%

Energy Saved

Through algorithmic optimisation techniques



Data Efficiency



Intelligent Filtering

Deduplication: removal of duplicate data to reduce storage and processing requirements.



Strategic Curation

Data curation: selection of relevant data to improve model quality whilst reducing computational overhead.

These approaches reduce storage and computation requirements whilst improving model relevance and accuracy.

Sustainable Infrastructure

Hardware & Energy Optimisation



Renewable Energy

Powering data centres with solar, wind, and other sustainable energy sources to reduce carbon footprint.



Optimised Cooling

Liquid cooling: liquid-based cooling systems. *Airflow management:* optimised air circulation for maximum efficiency.

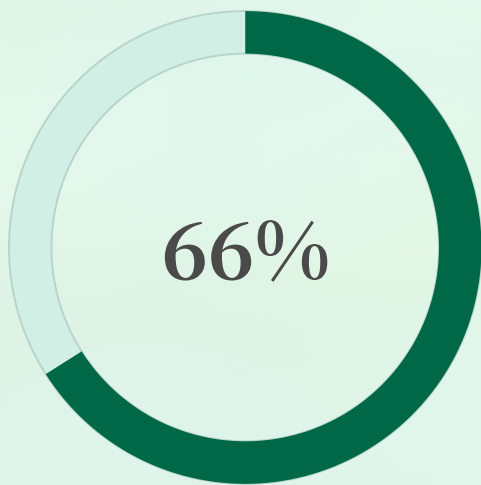


High-Efficiency Hardware

TPU / LPU (Large-scale Processing Unit): high-performance processors designed for energy-efficient AI workloads.

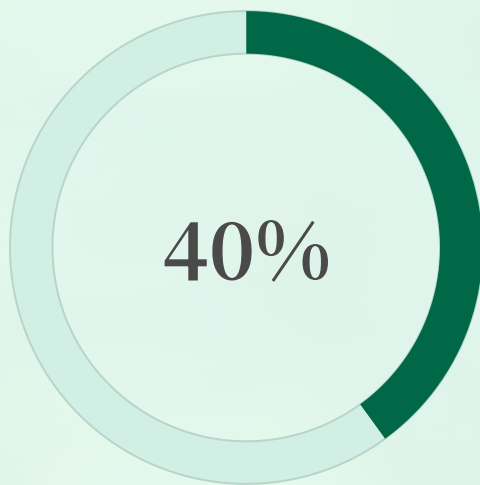


Google's Sustainability Leadership



Green Energy

Renewable energy powering operations

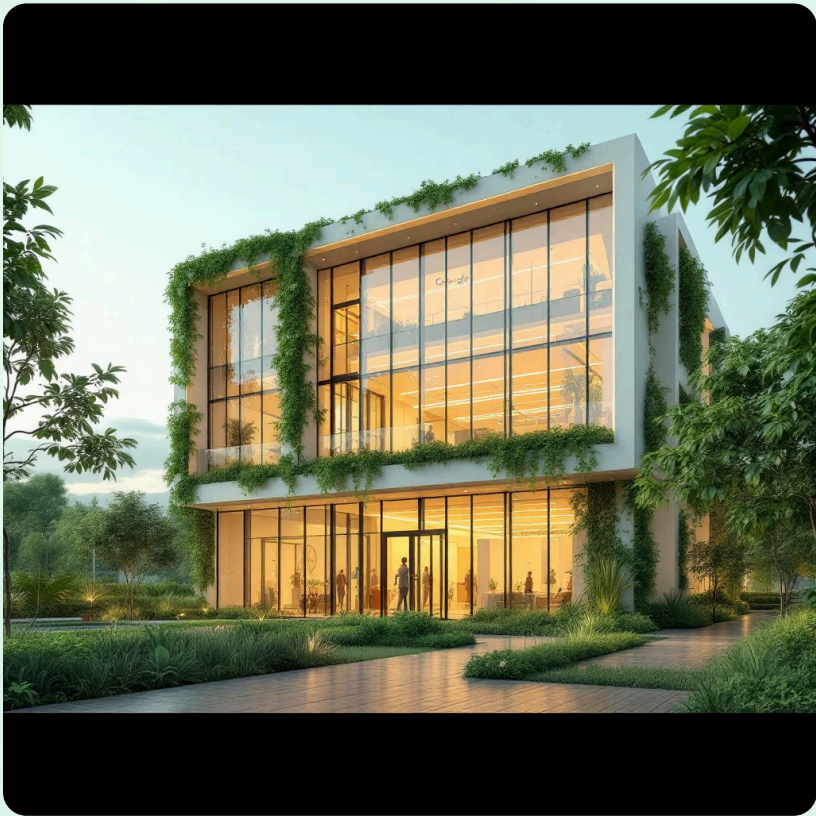


Cooling Reduction

Decrease in cooling energy consumption

Google has demonstrated that large-scale AI operations can be sustainable. Their comprehensive approach combines renewable energy adoption, advanced cooling technologies, and efficient hardware design.

Target: Carbon neutrality by 2030, setting the standard for the industry.



Roadmap to Green AI

"The future of AI relies on efficiency, not raw power."

Measure Carbon Impact

Integrate CO₂ tracking into **CI/CD** (continuous integration and deployment) pipelines to monitor environmental impact.

1

2

Optimise Models

Apply compression, pruning, quantization, and distillation techniques to reduce computational requirements.

Transition Infrastructure

Move towards **100% renewable** energy sources for all AI operations and data centres.

3

4

Efficiency-First Strategy

Prioritise efficient models over large ones, focusing on performance per watt rather than absolute capability.

Vision & Sources

📄 **Our Vision:** Reconciling innovation × performance × environmental responsibility to create sustainable AI systems that benefit society without compromising our planet.

Key Research & Reports

- Strubell et al., ACL 2019 — *Energy and Policy in Deep Learning*
- Schwartz et al., CACM 2020 — *Green AI*
- Patterson et al., Google 2022 — *Carbon Footprint of ML*
- IEA 2023 — *Data Centres & Energy*
- Google Sustainability Report 2023

Thank you for your attention

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